CSE 4830
Spring 2026 - Homework 2

Wednesday, Feb. 4, 2026

Due: 11:59 PM, Monday, Feb. 16, 2026

	Regular (Non-Honors)	**Honors**
Question 1	17 points	8 points
Question 2	17 points	12 points
Question 3	17 points	8 points
Question 4	17 points	8 points
Question 5	16 points	12 points
Question 6	16 points	8 points
Question 7	–	14 points
Question 8	–	14 points
Question 9	–	16 points
Total	**100**	**100**

1 Question 1

Consider a patient who undergoes two HIV diagnostic tests. Assume that the outcomes of the two tests are not independent. In particular, assume that either test on its own has a false positive rate of 10% and a false negative rate of 1%. That is, assume that:

$$P(D = 1 \mid H = 0) = 0.1 \quad \text{and} \quad P(D = 0 \mid H = 1) = 0.01.$$

Moreover, assume that for $H = 1$ (infected) the test outcomes are conditionally independent, i.e., that

$$P(D_1, D_2 \mid H = 1) = P(D_1 \mid H = 1)\, P(D_2 \mid H = 1),$$

but that for healthy patients the outcomes are coupled via

$$P(D_1 = D_2 = 1 \mid H = 0) = 0.02.$$

1. Work out the joint probability table for D_1 and D_2, given $H = 0$ based on the information you have so far.

2. Derive the probability that the patient is diseased ($H = 1$) after one test returns positive. You can assume the same baseline probability $P(H = 1) = 0.0015$.

3. Derive the probability that the patient is diseased ($H = 1$) after both tests return positive.

2 Question 2

1. Given a data matrix A with zero mean, show how the covariance matrix $C = \frac{1}{n-1}A^T A$ is formulated. Prove that C is a symmetric matrix.

   a) Calculate the eigenvalues and eigenvectors of C. Explain their significance in PCA.

   b) Prove that the eigenvectors of C are orthogonal.

2. In the context of PCA, describe the process of dimensionality reduction for a dataset.

   a) How do you select the number of principal components to keep?

   b) Formulate the mathematical transformation to project the data onto the selected principal components.

   c) Show how the original data can be approximately reconstructed from the reduced dataset.

3 Question 3

Why is Centering and Scaling the data important before performing PCA?

4 Question 4

Consider two lines, L_1 and L_2, on a Cartesian plane. It is known that the x-intercepts of L_1 and L_2 are 5 and -1, respectively. Additionally, L_1 has a y-intercept at 5, while L_2's y-intercept is at 1. Determine the coordinates of the intersection point for L_1 and L_2 in homogeneous coordinates.

5 Question 5

Using Google Colab, write a Python script to implement PCA on a dataset. Tasks to be performed:

- Load the Iris dataset (eg. `sklearn.datasets.load_iris()` (scikit-learn)).
- Standardize the dataset (mean = 0 and variance = 1 for each feature).
- Compute the covariance matrix of the standardized dataset.
- Calculate the eigenvalues and eigenvectors of the covariance matrix.
- Perform PCA to reduce the dataset to 2 principal components.
- Visualize the transformed dataset in a 2D plot.

Explain the impact of PCA on the dataset and the variance explained by the principal components.

6 Question 6

Using Google Colab, write a Python script to generate a simple image, similar to Figure 1, and perform various image transformations. Tasks to be performed:

- Generate a simple image using NumPy with the format **(300, 300, 3)**. This could be a plain colored background with basic shapes drawn on it.
- Apply a rigid transformation (combination of translation and rotation) to the image manually. Display the original and transformed images side by side.
- Convert the coordinates used in the rigid transformation between homogeneous and Cartesian forms.
- Apply a general 2D transformation to the image (e.g., shearing or scaling).
- Add shadows to the bottom of the transformed image.

Requirements:

- Limit the use of libraries to **OpenCV (cv2)**, **NumPy (np)**, and **Matplotlib (plt)** only. Do not use libraries for generating shadows, performing rigid transformations, or coordinate conversions. You need to write the code .
- Use **NumPy** to generate the initial simple image.
- Include comments in the code to explain each step of the process.

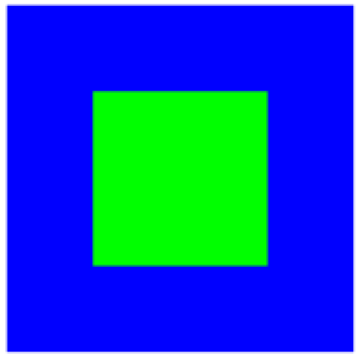

Figure 1: Example of simple image generation.

7 Question 7 - Only Honors Students

Prove that the principal components obtained from PCA are the eigenvectors of the covariance matrix of the data, and the corresponding eigenvalues represent the variance of the data along these components.

8 Question 8 - Only Honors Students

Using Google Colab, write a Python script to generate a complex image, similar to Figure 2, using NumPy and perform advanced image processing tasks. Tasks to be performed:

- Generate a complex image using NumPy with the format (300, 300, 3). The image should include multiple geometric shapes with varying colors and gradients.
- Implement a non-linear transformation (e.g., polar coordinate transformation) on the image manually. Visualize the original and transformed images.
- Develop an algorithm to identify and isolate a specific shape or color within the image.
- Apply a custom image filtering technique that you define (e.g., edge enhancement, custom blurring) without using pre-defined filters from libraries.

Requirements:

- Limit the use of libraries to **OpenCV (cv2)**, **NumPy (np)**, and **Matplotlib (plt)** only. Do not use libraries for generating shadows, performing rigid transformations, or coordinate conversions. These should be implemented manually.

- Use **NumPy** to generate the initial simple image.
- Include comments in the code to explain each step of the process.
- The image generation and processing tasks should demonstrate advanced understanding and creativity.

Figure 2: Example of complex image generation.

9 Question 9 - Only Honors Students

Prove that in PCA, the direction that maximizes the variance of the projected data is the eigenvector of the covariance matrix corresponding to the largest eigenvalue.

Submission

- Submit a Google Colab notebook with the complete code in *.ipynb* file with the outputs.
- Ensure the code is executable without errors and the outputs are visible in the notebook.